SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Engineering Conformance Specification

Scientific contract version:	**v0.1**
Document revision:	**r0.4**
Status:	**Unassessed conformance procedure**
Canonical science source:	**SCI-MAP v0.1 shared authority**
Target implementation:	**Not selected in this document**

No independent science. Every normative scientific clause and every prediction printed here is rendered from `SCI-MAP-v0.1_SHARED_AUTHORITY_r0.1.tex`. This document adds only evidence-recording, execution, traceability, and pass/fail procedure. If an engineering observation appears to require changing science, record a deviation and return it to scientific-owner review; do not edit the expected result in this view.

Contents

1 Purpose and conformance target

This specification tells a reviewer how to decide whether one exact candidate conforms to SCI-MAP v0.1. It does not select a candidate, inspect an implementation, execute validation, or assert a result. Before work begins, the reviewer freezes:

- repository and immutable candidate identity;
- build profile, compiler/runtime identities, configuration, and enabled parallel modes;
- exact input fixtures and independent expected-value generators;
- numerical comparison policy and simultaneous statistical error policy;
- required product list, representation paths, log policy, and completion criteria; and
- the shared-authority digest, contract version, this document revision, and owner-decision ledger revision.

Evidence from a different candidate, configuration, or science-authority digest is not silently pooled. Each rerun keeps its own identity and reason. Historical operational acceptance does not substitute for conformance evidence.

2 Evidence record

Each requirement record contains at least:

Field	Required content
Requirement	Stable `SCI-MAP-REQ-NNN` and shared-authority digest.
Candidate	Exact source/build identity and environment profile.
Observable	Public product, typed state, API boundary, metadata, diagnostic, or failure transition under test.
Fixture	Immutable input identity and independent expected-value calculation.
Comparison	Exact predicate or preregistered numerical/statistical bound.
Result	Pass, fail, blocked by named owner decision/dependency, or not applicable with owner-approved rationale. Skipped required evidence is not a pass.
Artifacts	Machine-readable outputs, logs, traces, plots/tables where needed, and digest-addressed provenance.
Reviewer	Independent reviewer identity, date, and deviation disposition.

Failures are retained as evidence. A tolerance, fixture, expected result, or applicability decision is not changed after viewing a failure without opening a new preregistered record.

3 Claim-layer gates

Passing one layer grants no stronger layer automatically.

Layer	Minimum engineering record	Prohibited inference
Algebraic contract	Owner-frozen shared authority and internally consistent independent expected values.	No claim that a candidate implements it.
Engineering conformance	Every applicable requirement passes on one exact candidate; no required skip, unresolved deviation, or false completion.	No claim that persisted WCS/response is scientifically faithful in real data.
Representation/response fidelity	Round-trip and injection evidence covers WCS, units, response classes, boundaries, and covariance representations with preregistered bounds.	No observational photometric, astrometric, or noise performance claim.
Observational performance	Independent references and repeatability establish the named amplitude, shape, position, covariance, or false-rate metric.	No production-readiness claim.
Production readiness	Operational profile, complete products, failure/log audit, dependency closure, release controls, and owner approval all pass.	No authority beyond the exact released profile and contract version.

An owner decision can block a claim without making the requirement disappear. Record the exact decision ID from the separate ledger and the conservative behavior observed.

4 Execution order and atomicity protocol

Run evidence in the following order so later evidence cannot mask an earlier contract failure:

1. Parse and freeze the realized PTC parent identity and source-declared availability/validity scope; reject a direct CAL input and a missing PTC product, while accepting an explicitly realized identity-like PTC product.
2. Parse and freeze identity, units, lifecycle, MAP policy (including the exact admitted value and unit status of every policy parameter), WCS, and required-product plan.
3. Separately verify PTC-owned source facts, ALIGN/AST-owned coordinate facts, and MAP-owned projection/coefficient/boundary/support/use conditions before arithmetic. Verify that VAL, if used, evaluates the named rule without authoring it and that no layer rescues a failed condition in another.
4. Compare exact scientific output-row identity and cardinality, plus one-pixel, one-hot, fractional, covariance, response, and eight-fact observation outputs, to independent expected values; retained storage rows do not count as scientific output rows.
5. Verify complete immutable observation-bundle construction.
6. Snapshot coadd-owned state; test every compatibility failure; require atomic nonmutation.
7. Exercise admitted centered-integer coaddition, exact coadd output-row identity, and companion placement.
8. Round-trip all required representations and reconstruct each row-selected operator from provenance.
9. Exercise honest response/covariance disclosure, including partial and unavailable states that leave the map valid, and verify versioned binding for later response, covariance, or corrected-map products.
10. Exercise consumer claim boundaries and immutable raw parentage without assuming a closed-world registry of all future scientific uses.
11. Repeat supported execution modes under the preregistered floating-point policy, while requiring exact discrete facts.

12. Only after deterministic gates pass, execute representation, external, observational, and operational evidence appropriate to the requested claim.

For every failure-injection test, capture pre-state, attempted input, emitted diagnostic, post-state, publication/cardinality state, and completion state. No partial-state comparison may omit a coadd companion merely because its payload was not the injected failure target.

5 Canonical requirement conformance clauses

The following clauses are generated from the shared authority. The first paragraph is the canonical science requirement; the second is its minimum conformance route. The result field intentionally remains unassessed.

SCI-MAP-REQ-001 – Versioned authority

Canonical clause: The scientific contract version shall be v0.1. The rendered document revision shall be recorded separately as r0.4; changing layout or exposition alone shall not be represented as a new scientific contract version.

Minimum conformance evidence: Inspect both artifacts and their machine-readable provenance for distinct contract-version and document-revision fields.

Result: UNASSESSED

SCI-MAP-REQ-002 – Capability and realized PTC input

Canonical clause: The ordinary input shall be a realized SCI-PTC MAP-facing product derived from calibrated ordinary-`xs` Stokes-I samples, with exact sample-by-detector membership and active signal unit `mJy/beam`. MAP shall have no CAL-to-MAP fallback: an identity-like PTC configuration remains a realized PTC product, and absence of the required PTC product means that MAP has no ordinary input.

Minimum conformance evidence: Exercise realized cleaned and identity-like PTC products, an attempted direct CAL input, a missing PTC product, and rejected stream, Stokes, and unit identities.

Result: UNASSESSED

SCI-MAP-REQ-003 – Explicit identity

Canonical clause: Every candidate contribution shall carry explicit observation, array, network when applicable, map-group, Stokes, input-column, and occurrence/product-scoped detector-acquisition identity. Container position, row equality, or a local key alone shall not establish scientific identity.

Minimum conformance evidence: Permute containers and rows while preserving or breaking explicit bindings; scientific products shall follow the binding and reject ambiguity.

Result: UNASSESSED

SCI-MAP-REQ-004 – Producer facts and MAP admission

Canonical clause: SCI-MAP shall consume without reinterpretation PTC-owned product identity, availability, producer-local validity, and MAP-facing coefficients, together with other producer-owned facts. MAP contribution shall require the realized PTC output to be available and every applicable MAP-owned projection, coefficient, boundary, finite-value, support, and use condition to pass. VAL may register or evaluate a named rule but shall not author either producer or MAP policy.

Minimum conformance evidence: Toggle each producer fact and MAP-owned condition independently; verify that MAP neither rescues an unavailable PTC output nor delegates policy authorship to VAL.

Result: UNASSESSED

SCI-MAP-REQ-005 – Coordinate and projection authority

Canonical clause: ALIGN/AST shall own sample coordinates, time/frame identity, astrometry, and coordinate validity. MAP shall own the target grid and its declared sample-to-map relation G_{pi}, projection class, normalization, extent, and boundary convention. MAP shall not choose or repair pointing, interpolate a missing coordinate, reinterpret coordinate validity, wrap, clamp, or invent an out-of-grid placement.

Minimum conformance evidence: Use centers, subpixel coordinates, boundaries, non-finite or invalid coordinates, and out-of-grid coordinates with independently calculated ALIGN/AST and MAP oracles.

Result: UNASSESSED

SCI-MAP-REQ-006 – Ordinary coefficient status

Canonical clause: Each admitted ordinary analysis coefficient shall be finite and strictly positive, with declared unit, normalization scope, lifecycle, applied factors, and statistical status. Zero is noncontributing; negative and non-finite ordinary coefficients are invalid.

Minimum conformance evidence: Inject positive, zero, negative, and non-finite coefficients and inspect contributions, diagnostics, and labels.

Result: UNASSESSED

SCI-MAP-REQ-007 – Exposure meaning

Canonical clause: Every exposure contribution shall be finite and nonnegative in a named unit and accounting convention. Detector-seconds shall not be labeled wall-clock integration time without an explicit multiplicity normalization.

Minimum conformance evidence: Use a fixture with unequal detector multiplicity and sample durations; compare both exposure facts to hand sums and inspect units.

Result: UNASSESSED

SCI-MAP-REQ-008 – Response information state

Canonical clause: MAP shall report the response information it actually provides, including meaning, domain, parentage, normalization, units, and limitations, or an explicit typed unavailable state. Missing or incomplete response shall not invalidate an otherwise valid map, fabricate an identity response, or prohibit later scientific analysis. A later simulated response or response-corrected map shall be a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Exercise complete, partial, and unavailable response states; verify honest claims, no sentinel or fabricated identity response, and versioned derivative-product binding.

Result: UNASSESSED

SCI-MAP-REQ-009 – Immutable plan and lifecycle

Canonical clause: The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

Minimum conformance evidence: Compare serialized states across two deliberately different observations and verify one-way provenance and complete replacement.

Result: UNASSESSED

SCI-MAP-REQ-010 – Contribution predicate

Canonical clause: For each pixel, the estimator contribution set shall require a realized PTC output available under the producer's exact declared identity and scope, plus finite signal, finite positive effective coefficient, admitted MAP projection and boundary, support/use policy, and every required MAP companion predicate before payload accumulation. No universal occurrence-to-product or product-to-occurrence failure hierarchy shall be inferred.

Minimum conformance evidence: Vary the PTC availability fact and one MAP predicate at a time, including different declared failure scopes, and compare the exact admitted set against independent producer and MAP oracles.

Result: UNASSESSED

SCI-MAP-REQ-011 – Ordinary accumulators

Canonical clause: On the fixed contribution set, SCI-MAP shall accumulate the pre-normalized signal numerator $N_p = \sum_i a_{pi} x_i$, normalization $Q_p = \sum_i a_{pi}$, and, when available, response numerator $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi}\omega_i > 0$.

Minimum conformance evidence: Use one-pixel, one-hot, and fractional hand calculations; compare accumulators before normalization.

Result: UNASSESSED

SCI-MAP-REQ-012 – Normalized publication

Canonical clause: The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows a finite $Q_p > 0$ gives $\widehat{x}_p = N_p/Q_p$ and an available response companion gives $\widehat{k}_p = K_p/Q_p$. Accumulators or storage slots may exist elsewhere, but no unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero sky.

Minimum conformance evidence: Exercise positive, exact-zero, near-zero finite, overflowed, and non-finite aggregates; compare the exact output-row set and its cardinality with an independent support oracle, and inspect any retained storage slot together with validity.

Result: UNASSESSED

SCI-MAP-REQ-013 – Accumulator interpretation

Canonical clause: The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

Minimum conformance evidence: Audit schema, metadata, API types, and labels for every numerator and normalization representation.

Result: UNASSESSED

SCI-MAP-REQ-014 – Constant preservation

Canonical clause: With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

Minimum conformance evidence: Map constant fixtures with uniform and unequal coefficients; compare exact output-row membership, then confirm that stronger response claims remain separately gated.

Result: UNASSESSED

SCI-MAP-REQ-015 – Projection-specific estimand

Canonical clause: One-hot assignment and fractional projection shall be identified explicitly. For fractional projection, normalized gridding shall not be mislabeled diagonal weighted least squares, and Q_p shall not be promoted to formal precision by analogy.

Minimum conformance evidence: Use a split-sample counterexample and compare normalized gridding, diagonal-WLS normalization, variance, and cross-pixel covariance.

Result: UNASSESSED

SCI-MAP-REQ-016 – Same operator for linear companions

Canonical clause: Signal, available response/kernel tracer, and declared linear realizations shall use the same admitted membership, geometric placement, and exact support-authorized output-row selection; retained exposure and coadd observation count shall use the same membership and placement wherever their definitions require them.

Minimum conformance evidence: Trace coefficient-by-coefficient membership, placement, and scientific output-row selection through signal and each declared linear companion.

Result: UNASSESSED

SCI-MAP-REQ-017 – Realized response claim

Canonical clause: A stored kernel may be called the realized response to template $\boldsymbol{\tau}$ only when its sample parent satisfies $\boldsymbol{k} = H\boldsymbol{\tau}$ and the support-restricted map product satisfies $\widehat{\boldsymbol{k}}_{\text{out}} = A_{\text{out}} H\boldsymbol{\tau} = R_{\text{out}}\boldsymbol{\tau}$ on exactly the authorized rows. Otherwise it shall be labeled only a tracer or unavailable.

Minimum conformance evidence: Inject a declared template and an intentionally mismatched tracer; compare exact response-row membership and accept the response label only for the proven identity.

Result: UNASSESSED

SCI-MAP-REQ-018 – Absolute-offset mode

Canonical clause: SCI-MAP v0.1 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

Minimum conformance evidence: Use a constant-offset fixture and inspect both the mapped value and recorded upstream response/null-mode state.

Result: UNASSESSED

SCI-MAP-REQ-019 – Conditional covariance

Canonical clause: When a declared PTC-domain covariance N_{PTC} is available for the asserted domain and fixed complete state Θ, MAP shall propagate it on the exact support-authorized row set as $C_{x,\mathrm{out}} = A_{\mathrm{out}} N_{\mathrm{PTC}} A_{\mathrm{out}}^{\mathsf{T}}$, retaining consequential cross-pixel terms. The equation does not require MAP to provide a complete covariance model when its input is absent or incomplete, and no zero-filled extension is scientific covariance.

Minimum conformance evidence: For a small explicit covariance, compare the exact row-selected propagation; separately verify truthful summarized, incomplete, and unavailable states without invalidating the signal map.

Result: UNASSESSED

SCI-MAP-REQ-020 – When normalization is precision

Canonical clause: The normalization may be called marginal inverse variance only when coefficients are calibrated true inverse variances, selected sample errors are independent for the asserted domain, and the projection satisfies $\sum_i \omega_i (G_{pi}^2 - G_{pi}) = 0$; one-hot assignment is the standard sufficient case.

Minimum conformance evidence: Test the authorized one-hot specialization and at least one fractional or correlated counterexample.

Result: UNASSESSED

SCI-MAP-REQ-021 – Formal diagonal weight

Canonical clause: When its assumptions hold on an authorized output row, the formal marginal inverse variance shall be $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \omega_i$. It shall remain distinct from Q_p, from the support-restricted full precision matrix $C_{x,\mathrm{out}}^{+}$, and from empirical statistical weight.

Minimum conformance evidence: Compute all four objects for a fractional fixture and verify exact row identity, distinct identities, and metadata.

Result: UNASSESSED

SCI-MAP-REQ-022 – Covariance and nuisance disclosure

Canonical clause: The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Inspect every representation state, prevent promotion of normalization, hits, or diagnostic weight to covariance, and verify derivative-product binding without rewriting the original claim.

Result: UNASSESSED

SCI-MAP-REQ-023 – Data-derived state

Canonical clause: If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

Minimum conformance evidence: Use metadata cases for fixed and data-derived state and verify that unconditional labels appear only with separately admitted evidence.

Result: UNASSESSED

SCI-MAP-REQ-024 – No significance promotion

Canonical clause: Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. Empirical calibration remains NOI-owned.

Minimum conformance evidence: Attempt each prohibited relabeling through all public representations and require rejection or an unmistakable non-significance label.

Result: UNASSESSED

SCI-MAP-REQ-025 – Eight distinct facts

Canonical clause: Every raw bundle shall separately represent geometric projected incidences, estimator-admitted contributions, admitted observation count for coadds, upstream-eligible exposure, retained exposure, numerical normalization support, science-policy support, and authoritative final raw science validity. No alias shall become authority for another fact.

Minimum conformance evidence: Construct cases in which each pair differs and verify eight separately typed identities.

Result: UNASSESSED

SCI-MAP-REQ-026 – Geometric incidences

Canonical clause: Geometric projected incidences shall count upstream-eligible entries whose declared projection touches a pixel, using an explicit fractional-projection counting convention, regardless of whether a positive ordinary coefficient is later admitted.

Minimum conformance evidence: Use zero-weight and fractionally split samples; compare the integer count with the declared touch convention.

Result: UNASSESSED

SCI-MAP-REQ-027 – Estimator contributions

Canonical clause: Estimator-admitted contributions shall count exactly the entries in the ordinary contribution set and shall exclude zero, negative, non-finite, flagged, and out-of-bounds candidates.

Minimum conformance evidence: Compare exact integer counts to an independently enumerated contribution set.

Result: UNASSESSED

SCI-MAP-REQ-028 – Admitted observation count

Canonical clause: For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

Minimum conformance evidence: Use one complete, one unsupported, and one rejected observation bundle and verify the per-pixel count.

Result: UNASSESSED

SCI-MAP-REQ-029 – Upstream-eligible exposure

Canonical clause: Upstream-eligible exposure shall sum the named exposure measure over the externally eligible projected population, before estimator-specific coefficient rejection, according to its declared exposure projection.

Minimum conformance evidence: Use eligible zero-weight entries and verify that upstream-eligible exposure changes without changing estimator contribution count.

Result: UNASSESSED

SCI-MAP-REQ-030 – Retained exposure

Canonical clause: Retained exposure shall sum the same named exposure measure over estimator-admitted membership and placement; it shall not be labeled precision.

Minimum conformance evidence: Use mixed eligible/admitted membership and compare retained exposure with hand sums and the normalization plane.

Result: UNASSESSED

SCI-MAP-REQ-031 – Adopted threshold selector

Canonical clause: For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p; for count $N > 0$, select zero-based $k = \lfloor(\lfloor 0.75N\rfloor + N)/2\rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population identity and exact dimensionless support-policy scalar $c =$ `coverage_cut` shall be recorded. The effective support policy shall explicitly admit that exact numerical c state before support resolution; otherwise the required product shall fail closed.

Minimum conformance evidence: Check empty and multiple population sizes against an independent integer-index oracle; inspect population identity and exact numerical c admission. Exercise negative, zero, non-finite, and greater-than-one c states without assuming their disposition; only an explicitly owner-authorized effective-policy state may proceed.

Result: UNASSESSED

SCI-MAP-REQ-032 – Separate support predicates

Canonical clause: Only after the effective support policy explicitly admits the exact value of the dimensionless scalar c, numerical normalization support shall require an explicit finite positive $Q_p \geq Q_\star c/10$, and science-policy support shall independently require an explicit finite positive $Q_p \geq Q_\star c$. This contract asserts no general numeric domain for c. Every non-authorized numerical state fails closed before a support row set is constructed. Neither support state shall establish final validity or precision.

Minimum conformance evidence: Exercise values below, at, and above both thresholds, including $N = 0$, and compare exact Boolean planes and output-row sets; verify fail-closed behavior for every numerical c state not explicitly admitted by the effective policy.

Result: UNASSESSED

SCI-MAP-REQ-033 – Authoritative raw validity

Canonical clause: One MAP raw science-validity state shall require membership in the exact support-authorized output-row set, admitted PTC parent identity, applicable MAP support predicates, finite normalized signal, and absence of an applicable declared-scope MAP product failure. Response or covariance may be explicitly incomplete or unavailable unless a particular MAP-authored claim requires it. A retained slot, finite payload, positive coefficient, exposure, hit count, support plane, or compatibility alias alone shall not make a pixel valid.

Minimum conformance evidence: Use truth-table fixtures that separate producer availability, MAP admission, response/covariance disclosure, and declared failure scopes; attempt every prohibited promotion.

Result: UNASSESSED

SCI-MAP-REQ-034 – Screen before arithmetic

Canonical clause: Invalid contributions shall be excluded before their numerical payload is evaluated. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

Minimum conformance evidence: Place NaN and Inf payloads behind false and true predicates and check contamination, diagnostics, and failure scope.

Result: UNASSESSED

SCI-MAP-REQ-035 – Exceptional contribution states

Canonical clause: Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

Minimum conformance evidence: Run one-cause-at-a-time fixtures and inspect cause-specific states plus unchanged neighboring pixels.

Result: UNASSESSED

SCI-MAP-REQ-036 – Complete observation bundle

Canonical clause: An observation map shall be one immutable ordered bundle containing signal, numerator, normalization identity, honest response state, honest uncertainty/covariance state, all eight facts where applicable, units, frame/WCS, shape/indexing, estimator, lifecycle, realized PTC parentage, and exact product identity. Explicitly incomplete or unavailable response/covariance states are complete disclosures, not automatically invalid bundles.

Minimum conformance evidence: Remove or alter each required identity or disclosure field in turn; distinguish a missing state declaration from a declared incomplete or unavailable state.

Result: UNASSESSED

SCI-MAP-REQ-037 – Atomic coadd admission

Canonical clause: A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

Minimum conformance evidence: Snapshot all coadd-owned state before injecting one incompatibility at a time and require byte- or field-exact nonmutation.

Result: UNASSESSED

SCI-MAP-REQ-038 – Centered integer common grid

Canonical clause: Observation and coadd shapes shall differ by nonnegative even row and column counts, with the corresponding integer offset mapping their reference pixels to the same world coordinate. Fractional shift, reprojection, interpolation, wrapping, implicit recentering, and a different sky grid are forbidden.

Minimum conformance evidence: Test every parity/sign case and WCS reference-pixel relation with an independent coordinate oracle.

Result: UNASSESSED

SCI-MAP-REQ-039 – Ordinary coadd estimator

Canonical clause: For atomically admitted bundles and finite positive observation coefficients u_{op}, the coadd shall accumulate $N_p^c = \sum_o u_{op}\widehat{x}_{op}$ and $Q_p^c = \sum_o u_{op}$, then publish $\widehat{s}_p = N_p^c/Q_p^c$ only for rows in the exact effective coadd support set $\mathcal{S}^c_{\mathrm{out}}$. No unsupported coadd row or zero-filled substitute is a scientific output.

Minimum conformance evidence: Use one observation and unequal positive coefficients; compare accumulators and the exact coadd output-row set with hand calculations and an independent support oracle.

Result: UNASSESSED

SCI-MAP-REQ-040 – Coadd companions

Canonical clause: Available response/kernel shall use the same atomic admission, centered-integer placement, and exact support-authorized output-row selection as coadd signal. Retained exposure and admitted observation count shall use the same admission and placement; the count increments once per admitted complete bundle at each applicable pixel.

Minimum conformance evidence: Use variable observation and coadd support plus offsets; compare signal/response row membership, companion placement, and exact count.

Result: UNASSESSED

SCI-MAP-REQ-041 – Coadd covariance and precision

Canonical clause: For fixed admission and support-restricted coadd operator B_{out}, conditional covariance shall be $C_{c,\mathrm{out}} = B_{\mathrm{out}} C_{\mathrm{obs}} B_{\mathrm{out}}^{\mathsf{T}}$, including consequential cross-observation blocks among authorized rows. Summed observation coefficients equal coadd precision only for independent unbiased maps of the same unit-gain estimand weighted by their true marginal inverse variances.

Minimum conformance evidence: Use independent and correlated two-observation covariance fixtures and compare exact row-selected propagation with any precision label.

Result: UNASSESSED

SCI-MAP-REQ-042 – Correlated-GLS boundary

Canonical clause: A correlated minimum-variance solution using $K_p^{-1}\mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

Minimum conformance evidence: Use positive- and negative-correlation covariance matrices that distinguish ordinary weighting from the GLS benchmark; require method identity to remain explicit.

Result: UNASSESSED

SCI-MAP-REQ-043 – Minimum product identity

Canonical clause: Every observation and coadd product shall identify role, external observation or coadd identity, array and group, network/detector occurrence when applicable, Stokes I, estimator, response and covariance disclosure/version, signal unit, frame, WCS, pixel shape/order, boundary policy, MAP policy identity, realized PTC parentage, and product identity. Scientific policy identity shall be durable and comparable without this contract prescribing JSON, hashes, or a sidecar format.

Minimum conformance evidence: Round-trip scientific identity through each required representation, detect deliberate collisions, and verify that implementation-specific serialization is not mistaken for scientific authority.

Result: UNASSESSED

SCI-MAP-REQ-044 – Frames and units

Canonical clause: Science products shall use their declared equatorial J2000 TAN WCS with degree-valued spatial axes; Pointing/OOF registered uses, if approved, shall use a declared AltAz tangent plane in arcseconds. Signal and a unit-bearing realized response carry `mJy/beam`; covariance carries its square and a conditionally valid formal weight its inverse square.

Minimum conformance evidence: Inspect coordinate and unit metadata and test that cross-frame or cross-unit coadd admission fails.

Result: UNASSESSED

SCI-MAP-REQ-045 – WCS and index persistence

Canonical clause: Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

Minimum conformance evidence: Round-trip representative centers, corners, axis signs, and index conversions; calculate maximum sky separation and exact orientation invariants.

Result: UNASSESSED

SCI-MAP-REQ-046 – Realized provenance

Canonical clause: Realized state shall record executed observations and groups, admitted contributions and bundles, accumulators, exact observation and coadd support-authorized output-row sets, all eight facts, response and covariance representation states, centered-integer offsets, completed required products, failures, and the exact requested/effective/resolved parents.

Minimum conformance evidence: Reconstruct each row-selected applied operator and its cardinality from provenance alone for a multi-observation fixture.

Result: UNASSESSED

SCI-MAP-REQ-047 – Failure scope and atomicity

Canonical clause: An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

Minimum conformance evidence: Inject each failure class at its boundary; inspect exact scope, diagnostics, nonmutation, and absence of a false completion marker.

Result: UNASSESSED

SCI-MAP-REQ-048 – Required observation products

Canonical clause: A complete observation bundle remains required when coaddition is enabled, and every product designated required by the effective v0.1 plan shall be published successfully before completion is asserted. Whether future abstract contracts require physical observation-map publication independent of an effective product policy remains an owner decision.

Minimum conformance evidence: Induce an observation-product publication failure during coadd operation and require overall failure with no misleading completion claim.

Result: UNASSESSED

SCI-MAP-REQ-049 – Immutable raw parent

Canonical clause: Downstream transformations shall preserve resolvable raw bundle identity and the authoritative raw validity state. They shall not rewrite raw validity or promote a raw-invalid pixel through a filtered or consumer-local validity alias.

Minimum conformance evidence: Create a downstream-local valid result over a raw-invalid parent and verify that both states remain distinct and the raw state immutable.

Result: UNASSESSED

SCI-MAP-REQ-050 – Complete consumer boundary

Canonical clause: No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include estimator/normalization identity, honest response and uncertainty/covariance disclosures, eight facts, additive-bias/null-mode state, units, frame/WCS, shape/indexing, grouping, lifecycle, parentage, dependency versions, and declared-scope failures. Each later analysis determines which available facts its own claim requires; this contract shall not impose a closed-world consumer/use registry.

Minimum conformance evidence: Exercise representative consumers with one claim-required fact missing at a time, and verify that analyses remain possible when they honestly accept a declared limitation.

Result: UNASSESSED

SCI-MAP-REQ-051 – Same fixed noise operator

Canonical clause: An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

Minimum conformance evidence: Propagate fixed deterministic vectors through signal and realization paths and compare row identities and operator coefficients, not filenames or shapes.

Result: UNASSESSED

SCI-MAP-REQ-052 – Method and claim separation

Canonical clause: This authority shall not be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

Minimum conformance evidence: Attempt each method/claim promotion and verify explicit package/method rejection plus independent claim-state fields.

Result: UNASSESSED

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6 Prediction execution matrix

Predictions are also imported unchanged. Engineering records attach fixtures, bounds, artifacts, and results; they do not rewrite the prediction.

SCI-MAP-PRED-001 – Uniform constant

Canonical prediction: For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized raw-valid output row; numerator and normalization scale together, and no unsupported scientific row is created.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-002 – Unequally weighted constant

Canonical prediction: For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-003 – One-pixel hand calculation

Canonical prediction: For an authorized output row with admitted values x_i and coefficients $a_i > 0$, the exact expected triplet is $(N, Q, \widehat{x}) = (\sum a_i x_i, \sum a_i, \sum a_i x_i / \sum a_i)$; for an unauthorized row only declared accumulators or storage state may remain, never a normalized scientific zero.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-004 – Fractional split

Canonical prediction: One sample split with $G_{p1} = G_{r1} = 1/2$ contributes to both pixels, creates cross-pixel covariance, and distinguishes normalized gridding from diagonal WLS. Two equal independent split samples give $Q = 1$, variance $1/2$, and $\phi = 2$ per affected pixel.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-005 – Delta input

Canonical prediction: A basis-pixel sky input produces the corresponding support-row-restricted column of $R_{\text{out}} = A_{\text{out}} H$, including projection and boundary truncation; it need not be a delta in the output map, and no value is predicted outside the authorized rows.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-006 – Point-source template

Canonical prediction: A declared point-source template $\boldsymbol{\tau}_b$ produces $R_{\text{out}} \boldsymbol{\tau}_b$ on exactly the support-authorized rows; peak, integrated, and fitted amplitudes are distinct functionals and need separate criteria.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-007 – Resolved and Fourier inputs

Canonical prediction: Resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{out}} \boldsymbol{\tau}$ on the authorized row set; a constant-input pass does not predict preservation of these modes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-008 – Variable coverage

Canonical prediction: Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-009 – Finite boundary

Canonical prediction: At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-010 – Coefficient states

Canonical prediction: Zero coefficients add no estimator contribution; low positive coefficients follow the adopted support policy; negative and non-finite ordinary coefficients are invalid and cause-specific.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-011 – Eligibility and non-finite states

Canonical prediction: Flagged, non-finite, and out-of-bounds candidates do not contaminate any accumulator because they are excluded before payload evaluation; zero or failed support produces no scientific output row rather than measured zero.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-012 – Threshold selector

Canonical prediction: For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor(\lfloor 0.75N\rfloor + N)/2\rfloor$; empty input yields threshold zero, but no pixel gains support without its own finite positive coefficient. The dimensionless identity of c is fixed; any exact numerical c value not explicitly admitted by the owner-authorized effective policy fails closed before a support row set is constructed, and no numeric domain is inferred.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-013 – Missing response

Canonical prediction: A missing response parent yields a typed unavailable response state, never a fabricated zero or identity response. The original MAP bundle remains usable for claims that do not require response; any later simulated response or corrected map has a new versioned identity bound to that bundle.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-014 – Atomic rejection

Canonical prediction: A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-015 – One-observation coadd

Canonical prediction: A coadd containing one admitted observation with positive coefficient reproduces that observation's signal and available response exactly on the corresponding centered-integer rows authorized by both the observation and coadd support policies; it creates no unsupported scientific row.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-016 – Unequal observation coefficients

Canonical prediction: On each support-authorized coadd row, two compatible observations produce the positive-coefficient weighted mean and the summed normalization; the result remains inside the convex hull of their finite values, while an unauthorized row has no normalized coadd value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-017 – Missing or invalid observation

Canonical prediction: An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-018 – Centered integer placement

Canonical prediction: Compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-019 – Cross-observation covariance

Canonical prediction: For two maps with variances σ_1^2, σ_2^2 and covariance c, the minimum-variance coefficients depend on c; summing marginal normalizations is not precision unless the required independence and calibration conditions hold. Covariance comparison uses the exact support-authorized observation and coadd rows.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-020 – Fixed noise operator

Canonical prediction: A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient A_{out} and B_{out} as the signal path; recomputed state exposes a named difference.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-021 – Sequential and parallel reduction

Canonical prediction: Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-022 – No false significance

Canonical prediction: A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-023 – No validity promotion

Canonical prediction: A downstream product may have its own local validity but cannot change an inherited raw-invalid pixel to raw-valid or sever its raw-parent identity.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-024 – No ordinary predicate for JINC

Canonical prediction: A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.1.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-025 – Aggregate and index failure

Canonical prediction: An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

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7 Mechanical completeness checks

Before a conformance verdict, automated checks shall establish:

- exactly one declaration of each stable requirement and prediction ID in the shared authority, with no gaps or duplicates;
- every requirement appears in this specification and in `CROSSWALK.md`; every prediction is linked to at least one requirement;
- each rendered PDF identifies contract v0.1 and its current document revision separately;
- no unresolved LaTeX reference, citation, overfull box, clipped glyph, broken table, or unreadable page remains;
- every required artifact reopens successfully, has the expected page count, and renders every page through an independent PDF renderer; and
- the owner-decision ledger and author-draft decision record contain no silent resolution of an open scientific question.

The conformance verdict is **pass** only when every applicable requirement has passing evidence on the exact candidate and all requested stronger claim layers have their own passing evidence. Otherwise the record is fail, blocked, or incomplete. This document itself records no verdict.

8 Required handoff contents

An engineering handoff contains the candidate and authority identities; requirement result table; prediction result table; exact commands and environment; independent expected-value artifacts; build/test/log outputs; representation and observational evidence where claimed; failure-injection snapshots; skipped/blocked items; deviations and owner dispositions; and an explicit statement for each claim layer. The handoff shall not collapse "conforms algebraically," "represents faithfully," "performs on sky," and "ready for production" into one status word.